

Model Question Paper

FIFTH SEMESTER DIPLOMA EXAMINATION IN AUTOMOBILE ENGINEERING

AUTOMOBILE DESIGN (4019)

PART - A

I Answer all questions. Each question carries 2 marks.

1. What is a link?
2. What is inversion?
3. What is a spline?
4. What is module?
5. Define pressure angle

(5x2 =10)

PART - B

II Answer any five questions. Each question carries 6 marks.

1. Explain scotch yoke mechanism.
2. Explain the general classification of bearing.
3. Explain the construction procedure of displacement diagram for simple Harmonic Motion of a follower.
4. Explain the terminology of a disc cam.
5. Explain step- by step procedure for the design of flange coupling.
6. A 400 mm diameter pulley is driven at 750 rpm by a flat belt with tight side tension of 300 N and slack side tension of 35 N. Determine the power transmitted by the belt.
7. A gear having 60 teeth meshes with a pinion having 40 teeth. Calculate the distance between the centers of the gears, if the circular pitch is 15.7mm.

PART – C

(Answer one full question from each unit. Each question carries 15 marks.)

UNIT – I

- III Compute the diameter of a solid shaft which has to transmit 16 KW power at 260 rpm. Ultimate shear stress for shaft material is 350N/mm^2 and factor of safety for design is 8. If a hollow shaft is replaces the solid shaft, find the inside and outside diameters if their ratio is 0.5. (15)

OR

- IV (a) Differentiate between lower pair and higher pair with examples. (5)
- (b) A solid shaft is to transmit a torque of 20KN-M. If the maximum shear stress induced in the shaft is not to exceed 55MPa, find the minimum diameter of the shaft. (10)

UNIT – II

- V Design and sketch a cast iron flange coupling for a shaft transmitting 15 KW at 200 rpm and having an allowable shear stress of 40 MPa. The working stress in the bolts should not exceed 30MPa. Assume that the same material is used for shaft and key and the crushing stress is 100Mpa. The maximum torque is 15% greater than the full load torque. The shear stress for cast iron is 14MPa. (15)

OR

- VI (a) A multiple disc clutch has 9 plates having contact surface of 25 mm inside radius and 50mm outside radius. Assume uniform wear and coefficient of friction as 0.2, find the maximum intensity of pressure between surface for transmitting 5 KW at 500 rpm. (10)
- (b) What are the important type of keys? Name them with simple sketches. (5)

UNIT – III

- VII A flat faced follower is operated by a uniform rotating cam. The follower is raised through a distance of 25mm in 120° rotation of the cam ,remains at rest for the next 30° and is lowered during further 120° rotation of the cam. The raising of the follower takes place with simple harmonic motion and the lowering with uniform velocity. The least radius of the cam is 25mm. (15)

OR

- VIII Design an aluminium alloy piston for a single acting four stroke engine for the following specifications:

Cylinder bore = 0.30m

Stroke =0.375m

Maximum gas pressure =8MPa.

Break mean effective pressure=1.15MPa.

Fuel consumption =0.22Kg/KW/Hr.

Speed =500rev/min.

Assume suitable data. (15)

UNIT – IV

- IX (a)Derive the expression for finding ratio of belt tensions. (9)

(b)Explain two types of gear trains used with suitable sketches. (6)

- X A four speed gear box is to have the following gear ratios 1.0, 1.5, 2.48 and 3.93. The centre distance between the lay shaft and main shaft is 73.12mm and the smallest pinion is to have at least 15 teeth with a module of 3.25mm. Find the number of teeth of various gears, find the exact gear ratio. (15)